

PHYS 357: Honours Quantum Physics 1

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August 31st, 2026

Abstract

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1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

2.1 The classical picture

2.1.1 Angular momentum

Definition 1 (Orbital angular momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (1)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr.$$

Definition 2 (Angular momentum of a rigid body). *For a rigid body¹ rotating about a fixed axis this is*

$$L = I_0\omega,$$

with I_0 the moment of inertia and ω the angular velocity.

Remark 1. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

2.1.2 Magnetic moment of a circulating charge

Definition 3 (Magnetic moment). *A magnetic moment is a measure of the net circulation of electric charge. It is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

¹A rigid body is one that does not deform under the action of external forces.

The magnetic moment of a point particle of mass m and charge q in a circular orbit of radius r and period T is

$$\mu = \frac{IA}{c} = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z,$$

where I is the current and A is the area enclosed by the orbit. A single circulating charge counts as a steady current $I = q/T = qv/2\pi r$ because it passes any given point on the loop once per period.

2.1.3 Gyromagnetic ratio

Definition 4 (Gyromagnetic ratio). *The gyromagnetic ratio γ is the constant of proportionality between the magnetic moment and the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}. \quad (2)$$

For the circulating charge above,

$$\gamma = \frac{q}{2mc}. \quad (3)$$

Remark 2. *This is a purely classical result, since the charge is treated as a point particle on a definite orbit with a definite radius and speed, and nothing is quantized. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

2.2 Spin

Definition 5 (Spin). *Spin $\vec{S}$ is an intrinsic angular momentum carried by a particle. It is a fixed property of the particle type, like charge or mass, and does not come from any motion through space. A stationary electron alone in empty space still has spin.*

In quantum mechanics the spin $\vec{S}$ replaces the orbital angular momentum $\vec{L}$ of the classical picture.

2.2.1 Intrinsic moment and the g-factor

The magnetic moment is still proportional to the angular momentum, but the gyromagnetic ratio picks up an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \quad \gamma = \frac{qg}{2mc}, \quad (4)$$

where g is the g-factor of the particle.

Definition 6 (g-factor). *The g-factor of a particle³ is the dimensionless number relating its magnetic moment to its spin angular momentum. It measures how much stronger or weaker the moment is than the classical value $\gamma = q/2mc$, which corresponds to $g = 1$.*

²Here c is in Gaussian units.

³Its value must be measured or derived from relativistic quantum theory.

2.2.2 Magnetons

Definition 7 (Planck constant). *The Planck constant h is the fundamental constant of quantum mechanics. It sets the scale at which quantization matters, and relates the energy of a photon to its frequency by $E = h\nu$. Its value is*

$$h \approx 6.626 \times 10^{-34} \text{ J s.}$$

It has units of $J \cdot s$, which is the same as angular momentum.

Definition 8 (Reduced Planck constant).

$$\hbar = \frac{h}{2\pi}. \quad (5)$$

It carries units of angular momentum, so $[\hbar] = [L_z] = \frac{kg \cdot m^2}{s}$. Its value is $\hbar = 1.055 \times 10^{-27} \text{ erg s} = 6.582 \times 10^{-16} \text{ eV s}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $q\hbar/2mc$ then carries the units and sets the natural size of the moment at a given mass scale.

Definition 9 (Bohr magneton). *The Bohr magneton is the natural unit of magnetic moment for an electron,*

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \quad (6)$$

Definition 10 (Nuclear magneton). *The nuclear magneton is the same combination with the proton mass, and is the natural unit for nucleons,*

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \quad (7)$$

Since $m_p/m_e \approx 1836$, we have $\mu_N \approx \mu_B/1836$.

Definition 11 (Nucleons). *The subatomic particles that make up the nucleus of an atom, specifically protons and neutrons.*

The electron moment is then written as

$$\vec{\mu}_e = g_e \mu_B \frac{\vec{S}}{\hbar}, \quad (8)$$

and similarly for the proton and neutron with μ_N in place of μ_B .

2.2.3 Values for the electron, proton and neutron

Particle	Charge q	Mass m	g -factor	Gyromagnetic ratio γ
Electron	$- e $	m_e	$g_e \approx 2$	$g_e \mu_B / \hbar$
Proton	$+ e $	m_p	$g_p \approx 5.59$	$g_p \mu_N / \hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n \mu_N / \hbar$

Table 1: Charges, masses, g -factors and gyromagnetic ratios.

Remark 3. The classical derivation gives $\mu = (q/2mc)L_z$, which vanishes when $q = 0$. The neutron is neutral yet has a nonzero moment, so its $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.

Remark 4. Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.

2.3 The Stern-Gerlach apparatus

Definition 12 (Stern-Gerlach apparatus). A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped to produce a strong field gradient along a chosen axis, here z . The gradient deflects each atom by an amount proportional to μ_z , and a screen records where it lands. Since $\vec{\mu} = \gamma\vec{S}$, the position on the screen is a direct measurement of S_z .

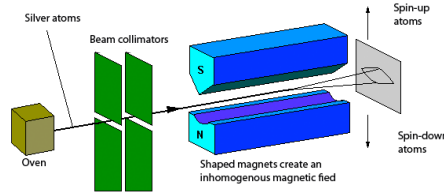


Figure 1: Illustration of the Stern-Gerlach apparatus.

2.3.1 Force in an inhomogeneous field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu} \cdot \vec{B}, \quad (9)$$

so the force on it is

$$\vec{F} = -\vec{\nabla}U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \quad F_z = -\frac{\partial U}{\partial z} = \mu_z \frac{\partial B_z}{\partial z}.$$

Definition 13 (Deflection). The displacement of an atom on the screen, perpendicular to the beam axis, produced by F_z acting over the length of the magnet. It is proportional to μ_z , so the pattern on the screen is a record of the moments in the beam.

Remark 5. A uniform field gives no force, only a torque. Thus, the apparatus needs a field gradient to deflect anything.

2.3.2 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin. For the electron,

$$\mu_z = -\frac{g_e |e|}{2m_e c} S_z,$$

and therefore

$$F_z = \mu_z \frac{\partial B_z}{\partial z} = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z,$$

where the last equality assumes $\partial B_z / \partial z < 0$.

Result 1.

$$F_z \propto S_z.$$

The deflection of an atom measures the z component of its spin directly.

2.3.3 The classical prediction

Spin is a vector,

$$\vec{S} = (S_x, S_y, S_z)^T, \quad S_z = |\vec{S}| \cos \theta. \quad (10)$$

Remark 6. *The spins leaving the oven point in random directions, so θ takes every value in $[0, \pi]$ across the beam and $S_z = |\vec{S}| \cos \theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Since $F_z \propto S_z$, each atom is deflected differently and the screen should show a continuous smear, but it shows two spots.*

2.3.4 The observed result

Result 2 (Quantization of spin). *The electron spin angular momentum is quantized:*

$$S_z = \pm \frac{\hbar}{2}. \quad (11)$$

The beam splits into two spots.

Definition 14 (Spin quantum number). *The spin quantum number s labels the magnitude of a particle's spin and is fixed by the particle type. It takes the values $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, and a measurement of S_z then gives one of $2s + 1$ outcomes,*

$$S_z = m\hbar, \quad m = -s, -s + 1, \dots, +s. \quad (12)$$

The electron has $s = \frac{1}{2}$, so $S_z = \pm \hbar/2$ and the beam splits in two. A spin-1 particle would split into three beams, $S_z = +\hbar, 0, -\hbar$.

Remark 7. *Everything in these notes is the $s = \frac{1}{2}$ case, so the state space is two-dimensional. Note also that S_z is a component, while the magnitude is $|\vec{S}| = \hbar\sqrt{s(s+1)}$, which for the electron is $\hbar\sqrt{3}/2$, not $\hbar/2$.*

Definition 15 (Basis state). One of a set of states $\{|j\rangle\}$ that are mutually orthonormal and span the space, so that every state is a unique linear combination of them. For spin- $\frac{1}{2}$ the two outcomes of any one measurement, $\{|\pm\hat{n}\rangle\}$, form a basis.

Definition 16 (Ket notation). A ket $|\cdot\rangle$ labels a quantum state as a vector in a vector space. For electron spin the two measurement outcomes give the basis states

$$|\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \quad |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2}.$$

The space is two-dimensional, so any spin state is a linear combination of the two.

Remark 8. From here on we follow Townsend and write $|+\hat{z}\rangle \equiv |\uparrow\rangle$ and $|-\hat{z}\rangle \equiv |\downarrow\rangle$. The same labels work for any axis: $|\pm\hat{x}\rangle$ are the states with $S_x = \pm\hbar/2$.

2.4 Sequential Stern-Gerlach experiments

Proposition 1. We write SG_z for a device whose gradient points along z , and SG_x for the same device rotated by 90° about the beam axis.

Definition 17 (Filter). Each device sends atoms into two beams. A filter is a device that blocks one of the beams.

Remark 9. A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation.

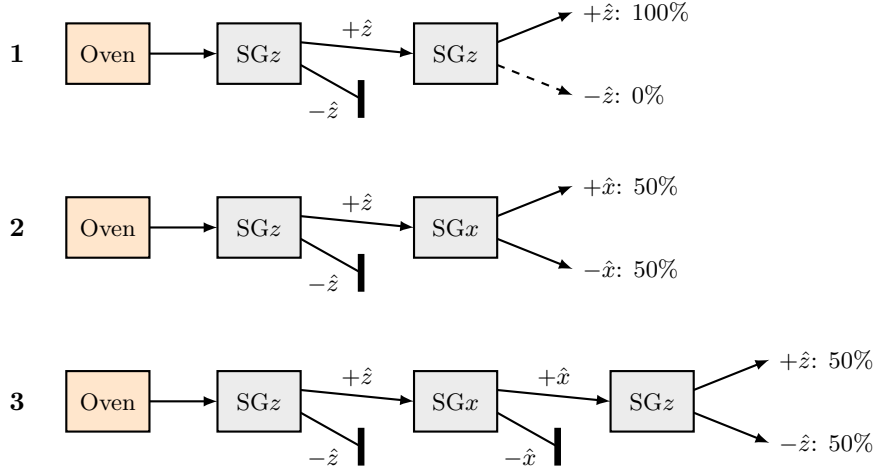


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device. Every oven starts with N_0 atoms.

2.4.1 Experiment 1: SG_z then SG_z

Filter the $+\hat{z}$ beam and send it into a second SG_z. Every atom comes out $+\hat{z}$, none come out $-\hat{z}$. The measurement is repeatable: once an atom is found with $S_z = +\hbar/2$, measuring S_z again gives the same answer. The atom leaves the first device in the state $|+\hat{z}\rangle$. So we end with $N_0/2$ atoms in $|+\hat{z}\rangle$.

2.4.2 Experiment 2: SG_z then SG_x

Filter the $+\hat{z}$ beam and send it into an SG_x. Half the atoms come out $+\hat{x}$ and half come out $-\hat{x}$. Knowing $S_z = +\hbar/2$ tells us nothing about S_x . We end up with $N_0/4$ in $|+\hat{x}\rangle$ and $N_0/4$ in $|-\hat{x}\rangle$.

2.4.3 Experiment 3: SG_z, SG_x, then SG_z

Filter $+\hat{z}$, then filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 into $+\hat{z}$ and $-\hat{z}$ at the end, even though every atom was filtered to $+\hat{z}$ at the start. The S_x measurement wiped out the earlier S_z information. Blocking $+\hat{x}$ instead of $-\hat{x}$ gives the same 50/50 result. We end with $N_0/8$ in $|+\hat{z}\rangle$ and $N_0/8$ in $|-\hat{z}\rangle$.

Result 3. *An atom with a definite value of S_x does not have a definite value of S_z . Measuring S_x destroys what we knew about S_z , so no state has definite values of both at once.*

2.4.4 Experiment 4: SG_z, modified SG_x, then SG_z

Definition 18 (Modified SG device). *An SG device that splits the beam according to S_n and then recombines the two beams, so atoms with $S_n = \pm\hbar/2$ emerge at the same position even though they took two different paths.*

Filter $+\hat{z}$, send it through a modified SG_x with neither path blocked, then measure S_z . Every atom comes out $+\hat{z}$: 100%, not the 50/50 of Experiment 3. All $N_0/2$ atoms end in $|+\hat{z}\rangle$.

Result 4. *The modified SG_x cannot be sending $N_0/4$ through as $|+\hat{x}\rangle$ and $N_0/4$ as $|-\hat{x}\rangle$: Experiment 3 showed each of those alone gives 50/50 at the end. Instead the two paths interfere, as for water or light waves, cancelling the possibility of $-\hat{z}$. No path can be assigned to an atom.*

3 Quantum States as Vectors

3.1 The vector space of spin states

3.1.1 Superposition

Definition 19 (Superposition). *A general spin state is a linear combination of the basis kets,*

$$|\psi\rangle = c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle, \quad c_{\pm} \in \mathbb{C}. \quad (13)$$

The pair $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$ is a basis, and the coefficients $c_{\pm}$ are complex numbers.

3.1.2 Bras and the inner product

Definition 20 (Bra). *Every ket $|\psi\rangle$ has a matching bra $\langle\psi|$. Putting a bra in front of a ket gives a complex number, the inner product*

$$\langle\phi| |\psi\rangle \equiv \langle\phi|\psi\rangle \in \mathbb{C}.$$

Definition 21 (Orthonormal basis). *The basis kets are orthonormal:*

$$\langle+\hat{z}| +\hat{z}\rangle = \langle-\hat{z}| -\hat{z}\rangle = 1, \quad \langle+\hat{z}| -\hat{z}\rangle = \langle-\hat{z}| +\hat{z}\rangle = 0. \quad (14)$$

3.1.3 Completeness

Definition 22 (Projector). *The operator $P_j = |j\rangle\langle j|$ built from a basis state. Acting on $|\psi\rangle = \sum_k c_k |k\rangle$ it keeps only the $|j\rangle$ part,*

$$P_j |\psi\rangle = c_j |j\rangle,$$

since $\langle j|k\rangle = \delta_{jk}$.

Proposition 2 (Completeness relation). *Since any state can be expanded in the basis $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$, the sum of the projectors onto the basis states is the identity:*

$$|+\hat{z}\rangle\langle+\hat{z}| + |-\hat{z}\rangle\langle-\hat{z}| = \mathbf{1}. \quad (15)$$

Inserting the identity in front of any state gives its expansion directly:

$$|\psi\rangle = (|+\hat{z}\rangle\langle+\hat{z}| + |-\hat{z}\rangle\langle-\hat{z}|) |\psi\rangle = \langle+\hat{z}|\psi\rangle |+\hat{z}\rangle + \langle-\hat{z}|\psi\rangle |-\hat{z}\rangle, \quad (16)$$

so the coefficients $c_{\pm}$ of the superposition are the inner products $\langle\pm\hat{z}|\psi\rangle$.

3.2 Extracting probabilities

3.2.1 Probability amplitudes

Proposition 3. *To pull out c_+ , act on $|\psi\rangle$ with $\langle+\hat{z}|$:*

$$\langle+\hat{z}|\psi\rangle = c_+ \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + c_- \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = c_+. \quad (17)$$

The same steps with $\langle-\hat{z}|$ give c_- . So

$$c_{\pm} = \langle\pm\hat{z}|\psi\rangle. \quad (18)$$

Definition 23 (Probability amplitude). *The coefficient $c_{\pm} = \langle\pm\hat{z}|\psi\rangle$ is the probability amplitude for measuring $S_z = \pm\hbar/2$. The probability itself is the modulus squared,*

$$P(S_z = \pm\frac{\hbar}{2}) = |c_{\pm}|^2 = |\langle\pm\hat{z}|\psi\rangle|^2. \quad (19)$$

Principle 1 (Born rule). *The probability that a system in the state $|\psi\rangle$ is found in the state $|\phi\rangle$ is*

$$P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 \in [0, 1]. \quad (20)$$

Remark 10. *Why is the Born rule true? There is no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."*

3.2.2 Normalization

Proposition 4. *A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, so*

$$P(\psi|\psi) = |\langle\psi|\psi\rangle|^2 = 1. \quad (21)$$

To find what a bra equals to, write it with unknown coefficients, $\langle\psi| = c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|$. Then

$$\langle\psi|\psi\rangle = (c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|) (c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle) = c'_+ c_+ + c'_- c_-,$$

since the cross terms vanish by orthonormality. The two outcomes $S_z = \pm\hbar/2$ must have probabilities $|c_+|^2$ and $|c_-|^2$ that add to 1. For $\langle\psi|\psi\rangle$ to be that sum, each product $c'_\pm c_\pm$ must be real and equal to $|c_\pm|^2$. This forces $c'_\pm = c_\pm^*$.

Result 5. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle\psi| = c_+^* \langle+\hat{z}| + c_-^* \langle-\hat{z}|. \quad (22)$$

In particular $\langle\psi|\pm\hat{z}\rangle = c_\pm^ = \langle\pm\hat{z}|\psi\rangle^*$. More generally, $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$.*

Definition 24 (Normalization condition).

$$\langle\psi|\psi\rangle = \langle\psi|+\hat{z}\rangle \langle+\hat{z}|\psi\rangle + \langle\psi|-\hat{z}\rangle \langle-\hat{z}|\psi\rangle = |c_+|^2 + |c_-|^2 = 1. \quad (23)$$

In words, $P(S_z = +\hbar/2) + P(S_z = -\hbar/2) = 1$: the atom is always found in one of the two beams.

Example. *Take*

$$|\psi\rangle = \frac{1}{\sqrt{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle.$$

The amplitude for $S_z = +\hbar/2$ is

$$\langle+\hat{z}|\psi\rangle = \frac{1}{\sqrt{3}} \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + i\sqrt{\frac{2}{3}} \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = \frac{1}{\sqrt{3}},$$

so $P(S_z = +\hbar/2) = 1/3$. In the same way $\langle-\hat{z}|\psi\rangle = i\sqrt{2/3}$, so $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. The two add to 1, so the state is normalized.

3.2.3 Global phase

Proposition 5. *Take two kets that differ only by an overall phase, $|\psi\rangle = e^{i\alpha} |\psi'\rangle$ with $\alpha \in \mathbb{R}$. For any state $|\phi\rangle$,*

$$P(\phi | \psi) = |\langle \phi | \psi \rangle|^2 = |e^{i\alpha} \langle \phi | \psi' \rangle|^2 = |\langle \phi | \psi' \rangle|^2 = P(\phi | \psi'), \quad (24)$$

since $|e^{i\alpha}| = 1$.

Result 6. $|\psi\rangle$ and $e^{i\alpha} |\psi\rangle$ give the same probabilities for every possible measurement, so they describe the same physical state. An overall (global) phase has no physical meaning and can be dropped.

Remark 11. A relative phase is different: it sits between the two terms of a superposition, so it cannot be pulled out front and dropped. It has physical consequences, as the next subsection shows.

3.3 Fixing the basis states

3.3.1 The state $|+\hat{x}\rangle$

In Experiment 3, atoms in $|+\hat{x}\rangle$ enter an SG z and split 50/50:

$$P(S_z = \pm \frac{\hbar}{2}) = |\langle \pm \hat{z} | +\hat{x} \rangle|^2 = \frac{1}{2}.$$

So both amplitudes have modulus $1/\sqrt{2}$, and each can carry its own phase:

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} e^{i\delta_+} |+\hat{z}\rangle + \frac{1}{\sqrt{2}} e^{i\delta_-} |-\hat{z}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle), \quad \delta = \delta_- - \delta_+,$$

with $\delta_{\pm} \in \mathbb{R}$. Dropping the global phase $e^{i\delta_+}$ leaves

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle).$$

Remark 12. The relative phase δ survives, and Experiment 3 alone does not fix it. It is pinned down by combining more measurements (for example along y) with a phase convention.

3.3.2 Experiment 5: measuring S_y

Definition 25 (Cyclic permutation). *A cyclic permutation of the axes relabels them as $x \rightarrow y$, $y \rightarrow z$, $z \rightarrow x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes under it.*

Let Experiment 5 be a cyclic permutation of Experiment 3, SG x becomes SG z and SG z becomes SG y . Now the process is as follows: filter $+\hat{z}$, then measure S_y .

Proposition 6. *By symmetry it must split 50/50 as well, so every cross-axis overlap has the same modulus:*

$$|\langle +\hat{x} | +\hat{z} \rangle|^2 = |\langle +\hat{y} | +\hat{z} \rangle|^2 = |\langle -\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2}. \quad (25)$$

The same argument as for $|+\hat{x}\rangle$ then gives

$$|+\hat{y}\rangle = \frac{e^{i\gamma_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\gamma} |-\hat{z}\rangle), \quad \gamma = \gamma_- - \gamma_+.$$

3.3.3 The state $|+\hat{y}\rangle$ and the phase convention

Proposition 7. *Dropping the global phases, $\gamma_+ = \delta_+ = 0$, we obtain equations for x and y in terms of z :*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\delta} |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\gamma} |-\hat{z}\rangle).$$

So using orthonormality of the $\hat{z}$ basis,

$$\langle +\hat{x} | +\hat{y} \rangle = \frac{1}{2} (1 + e^{i(\gamma-\delta)}), \quad |\langle +\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2} (1 + \cos(\gamma - \delta)).$$

Requiring this to equal $1/2$ forces $\cos(\gamma - \delta) = 0$, that is $\gamma - \delta = \pm\pi/2$. The remaining freedom is convention.

Result 7 (Phase convention). *Take $\delta = 0$ and $\gamma = +\pi/2$. Then*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm i |-\hat{z}\rangle). \quad (26)$$

The bras are then

$$\langle \pm\hat{x} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \pm \langle -\hat{z} |), \quad \langle \pm\hat{y} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \mp i \langle -\hat{z} |). \quad (27)$$

Remark 13. *Complex coefficients are required, not just allowed. With real coefficients the relative phases could only be 0 or π , so $\gamma - \delta = \pm\pi/2$ would be impossible.*

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle$, compute $\langle S_y \rangle$.*

The amplitudes are

$$\begin{aligned} \langle +\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle +\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle +\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (-i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + \sqrt{\frac{2}{3}} \right), \\ \langle -\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle -\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle -\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} - \sqrt{\frac{2}{3}} \right). \end{aligned}$$

Computing the probabilities

$$P(S_y = +\frac{\hbar}{2}) = |\langle +\hat{y} | \psi \rangle|^2 = \frac{1}{2} + \frac{\sqrt{2}}{3}, \quad P(S_y = -\frac{\hbar}{2}) = |\langle -\hat{y} | \psi \rangle|^2 = \frac{1}{2} - \frac{\sqrt{2}}{3},$$

which add to 1 as they must. Then

$$\langle S_y \rangle = \frac{\hbar}{2} \left[\left(\frac{1}{2} + \frac{\sqrt{2}}{3} \right) - \left(\frac{1}{2} - \frac{\sqrt{2}}{3} \right) \right] = \frac{\sqrt{2}}{3} \hbar.$$

The factor i that did nothing for S_z matters here: it is a relative phase, and it is exactly what tilts the spin toward $+\hat{y}$.

3.4 Statistics of measurements

3.4.1 Expectation value

Definition 26 (Expectation value). *The expectation value of S_z in the state $|\psi\rangle$ is the average outcome weighted by the probabilities,*

$$\langle S_z \rangle = \left(\frac{\hbar}{2} \right) |\langle +\hat{z} | \psi \rangle|^2 + \left(-\frac{\hbar}{2} \right) |\langle -\hat{z} | \psi \rangle|^2. \quad (28)$$

Remark 14. For N measurements on identically prepared atoms, with N_+ giving $+\hbar/2$ and N_- giving $-\hbar/2$, the measured average is

$$\langle S_z \rangle_N = \frac{\hbar}{2} \frac{N_+ - N_-}{N}, \quad (29)$$

which tends to $\langle S_z \rangle$ as $N \rightarrow \infty$.

Remark 15 (Shot noise). *For a finite number N of measurements, the sample average deviates from the expectation value by*

$$|\langle S_z \rangle_N - \langle S_z \rangle| \sim \frac{\Delta S_z}{\sqrt{N}}. \quad (30)$$

This spread is intrinsic, not a flaw of the apparatus: it appears whenever a state with a definite value of one observable (e.g. S_x for $|+\hat{x}\rangle$) is measured for an incompatible observable (e.g. S_z).

3.4.2 Uncertainty

Definition 27 (Uncertainty). *The uncertainty ΔS_z is the standard deviation of the measurement outcomes,*

$$(\Delta S_z)^2 = \langle (S_z - \langle S_z \rangle)^2 \rangle = \langle S_z^2 \rangle - \langle S_z \rangle^2, \quad (31)$$

where

$$\langle S_z^2 \rangle = \left(\frac{\hbar}{2} \right)^2 |\langle +\hat{z} | \psi \rangle|^2 + \left(-\frac{\hbar}{2} \right)^2 |\langle -\hat{z} | \psi \rangle|^2. \quad (32)$$

Example. For $|+\hat{x}\rangle$, both probabilities are $1/2$, so

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{2} - \frac{\hbar}{2} \cdot \frac{1}{2} = 0, \quad \langle S_z^2 \rangle = \left(\frac{\hbar}{2} \right)^2 \cdot \frac{1}{2} + \left(-\frac{\hbar}{2} \right)^2 \cdot \frac{1}{2} = \frac{\hbar^2}{4},$$

and therefore $\Delta S_z = \hbar/2$. The average is zero but every single measurement gives $\pm\hbar/2$, so the spread is maximal.

Example. For $|\psi\rangle = \sqrt{\frac{1}{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, the probabilities are $1/3$ and $2/3$, so

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{3} - \frac{\hbar}{2} \cdot \frac{2}{3} = -\frac{\hbar}{6}, \quad \langle S_z^2 \rangle = \frac{\hbar^2}{4} \left(\frac{1}{3} + \frac{2}{3} \right) = \frac{\hbar^2}{4}.$$

Then

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{9\hbar^2 - \hbar^2}{36} = \frac{2}{9}\hbar^2, \quad \Delta S_z = \frac{\sqrt{2}}{3}\hbar.$$

Remark 16. For spin- $\frac{1}{2}$, $\langle S_z^2 \rangle$ is always equal to $\frac{\hbar^2}{4}$, since both outcomes square to the same number and the probabilities sum to 1.

3.5 Generalization to n -level systems

3.5.1 Orthonormality and completeness

A spin- $\frac{1}{2}$ particle is a two-level system, and each of $\{|\pm\hat{x}\rangle\}$, $\{|\pm\hat{y}\rangle\}$, $\{|\pm\hat{z}\rangle\}$ is a basis. An n -level system has a basis $\{|j\rangle\}$, $j = 1, \dots, n$.

Remark 17. The index j labels the outcomes of one chosen observable, not the coordinate axes. Picking S_x , S_y or S_z picks the basis; within that basis $j = 1, 2$ labels its two states, for example $|a_1\rangle = |+\hat{x}\rangle$ and $|a_2\rangle = |-\hat{x}\rangle$. A spin-1 particle has $n = 3$ outcomes, and an n -level atom has $j = 1, \dots, n$.

Definition 28 (Orthonormal and complete basis). The basis is orthonormal if

$$\langle i|j\rangle = \delta_{ij},$$

and complete if every state can be expanded in it,

$$|\psi\rangle = \sum_j c_j |j\rangle, \quad c_j = \langle j|\psi\rangle, \quad \langle\psi| = \sum_j c_j^* \langle j|. \quad (33)$$

Completeness is equivalent to

$$\mathbb{1} = \sum_{j=1}^n |j\rangle \langle j|. \quad (34)$$

Example. For $n = 2$, $\mathbb{1} = |+\hat{n}\rangle \langle +\hat{n}| + |-\hat{n}\rangle \langle -\hat{n}|$ for any $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$, so

$$|\psi\rangle = \langle +\hat{x}|\psi\rangle |+\hat{x}\rangle + \langle -\hat{x}|\psi\rangle |-\hat{x}\rangle = \langle +\hat{y}|\psi\rangle |+\hat{y}\rangle + \langle -\hat{y}|\psi\rangle |-\hat{y}\rangle.$$

3.5.2 Observables and the Born rule

Definition 29 (Observable). An observable A has outcomes a_j with states $|a_j\rangle$ forming a complete orthonormal set. For spin- $\frac{1}{2}$, A could be S_x , S_y or S_z , with outcomes $\pm\hbar/2$. By the Born rule,

$$P(A = a_j) = |\langle a_j|\psi\rangle|^2. \quad (35)$$

Proposition 8. *Normalization $\langle\psi|\psi\rangle = 1$ is the statement that the probabilities sum to 1.*

Proof.

$$\langle\psi|\psi\rangle = \left(\sum_i c_i^* \langle a_i|\right) \left(\sum_j c_j |a_j\rangle\right) = \sum_{i,j} c_i^* c_j \delta_{ij} = \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.$$

Equivalently, inserting the identity, $\sum_j \langle\psi|a_j\rangle \langle a_j|\psi\rangle = \langle\psi| \left(\sum_j |a_j\rangle \langle a_j|\right) |\psi\rangle = \langle\psi|\psi\rangle$. $\square$

3.5.3 Expectation value as an operator

Proposition 9.

$$\langle A \rangle = \sum_j a_j |\langle a_j|\psi\rangle|^2 = \sum_j \langle\psi|a_j\rangle a_j \langle a_j|\psi\rangle = \langle\psi| \hat{A} |\psi\rangle, \quad \hat{A} = \sum_j a_j |a_j\rangle \langle a_j|. \quad (36)$$

Definition 30 (Uncertainty of an observable).

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2, \quad \langle A^2 \rangle = \langle\psi| \hat{A}^2 |\psi\rangle = \sum_j a_j^2 |c_j|^2. \quad (37)$$

4 Rotation of Basis States and Matrix Mechanics

4.1 States as column vectors

Definition 31 (Representation of a state). *In the S_z basis, a ket is represented by the column of its components and a bra by the conjugate row, the adjoint:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \quad \langle\psi| \xrightarrow{S_z} \psi^\dagger = (c_+^* \quad c_-^*), \quad (38)$$

with $c_\pm = \langle \pm \hat{z} | \psi \rangle$. Then

$$\langle\psi|\psi\rangle = \psi^\dagger \psi = |c_+|^2 + |c_-|^2. \quad (39)$$

Remark 18. Write $|\psi\rangle \xrightarrow{S_z} \psi$, not $|\psi\rangle = \psi$. The ket is basis-independent; the column depends on the basis, just like $\vec{E}$ versus its components.

Example.

$$|\pm \hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad |\pm \hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

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Example (Other bases). In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x}|\psi\rangle \\ \langle -\hat{x}|\psi\rangle \end{pmatrix}$, and similarly for S_y . So the same state has different columns:

$$|+\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x}|+\hat{z}\rangle \\ \langle -\hat{x}|+\hat{z}\rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_y} \begin{pmatrix} \langle +\hat{y}|+\hat{z}\rangle \\ \langle -\hat{y}|+\hat{z}\rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

4.2 Operators as matrices

Definition 32 (Projection operator). $P_{\pm\hat{n}} = |\pm\hat{n}\rangle\langle\pm\hat{n}|$, so that $\mathbb{1} = P_{+\hat{n}} + P_{-\hat{n}}$ and $P_{+\hat{z}}|\psi\rangle = c_+|+\hat{z}\rangle$.

Definition 33 (Matrix of an operator). In a basis $\{|j\rangle\}$, an operator A is represented by the matrix

$$A_{ij} = \langle i|A|j\rangle. \quad (40)$$

Inserting the identity, $\langle k|A|\psi\rangle = \sum_j \langle k|A|j\rangle\langle j|\psi\rangle$, so $A|\psi\rangle$ is represented by the matrix product $A\psi$.

Example.

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix} \langle +\hat{z}|P_{+\hat{z}}|+\hat{z}\rangle & \langle +\hat{z}|P_{+\hat{z}}|-\hat{z}\rangle \\ \langle -\hat{z}|P_{+\hat{z}}|+\hat{z}\rangle & \langle -\hat{z}|P_{+\hat{z}}|-\hat{z}\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} c_+ \\ c_- \end{pmatrix} = \begin{pmatrix} c_+ \\ 0 \end{pmatrix}.$$

Proposition 10 (Properties of projectors). A projector satisfies $P_{\hat{n}}^2 = P_{\hat{n}}$, since $(|\hat{n}\rangle\langle\hat{n}|)(|\hat{n}\rangle\langle\hat{n}|) = |\hat{n}\rangle\langle\hat{n}|\hat{n}\rangle\langle\hat{n}| = |\hat{n}\rangle\langle\hat{n}|$. In the S_z basis,

$$P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbb{1} = P_{+\hat{z}} + P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (41)$$

Definition 34 (Adjoint). Writing $A = \mathbb{1}A\mathbb{1} = \sum_{i,j} A_{ij} |i\rangle\langle j|$, the adjoint is

$$A^\dagger = \sum_{i,j} A_{ij}^* |j\rangle\langle i|, \quad (A^\dagger)_{ij} = A_{ji}^*. \quad (42)$$

If $|\psi'\rangle = A|\psi\rangle$, then $\langle\psi'| = \langle\psi|A^\dagger$.

4.3 The rotation operator

Definition 35 (Rotation operator). $R(\theta\hat{n})$ rotates a state by θ about $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})|\psi\rangle$. For example, $R(\frac{\pi}{2}\hat{y})$ takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ (up to phase), so

$$R\left(\frac{\pi}{2}\hat{y}\right)(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+|+\hat{x}\rangle + c_-|-\hat{x}\rangle. \quad (43)$$

Remark 19. $R(\theta\hat{n})$ is a linear operator: $R(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+R|+\hat{z}\rangle + c_-R|-\hat{z}\rangle$, which is what the equation above uses.

Remark 20. The bra of $|\psi'\rangle$ is $\langle\psi|R^\dagger$, not $\langle\psi|R$. With the wrong choice, $|\psi\rangle = |+\hat{z}\rangle$ and $\theta = \pi/2$ would give $\langle\psi'|\psi'\rangle = \langle+\hat{z}|R(\pi\hat{y})|+\hat{z}\rangle = \langle+\hat{z}|-\hat{z}\rangle = 0$ instead of 1.

Theorem 1. *Rotation operators are unitary:*

$$R^\dagger R = R R^\dagger = \mathbb{1}. \quad (44)$$

Proof. A rotation must preserve normalization: $\langle \psi' | \psi' \rangle = \langle \psi | R^\dagger R | \psi \rangle = 1$ for every $|\psi\rangle$ and every $R(\theta \hat{n})$, which forces $R^\dagger R = \mathbb{1}$. $\square$

Remark 21. *Unitary operators map physical (normalized) states to physical states. If a unitary U satisfies $U |\psi\rangle = c |\psi\rangle$, then $|c| = 1$, so $c = e^{i\varphi}$ is just a global phase and $U |\psi\rangle$ describes the same physical state as $|\psi\rangle$.*

A Appendix

B Useful Links

Stern-Gerlach experiment: <https://www.youtube.com/shorts/QG7kJXxN1Kk>